PRODUCT REQUIREMENT DOCUMENT (PRD)

ScamBreaker

UMHackathon 2026

April 2026

Table of Contents

Project Name: ScamBreaker
Version: 1.0
AI Model: ZhipuAI GLM (Z AI Model) - GLM-5.1 / GLM-4.6v / GLM-4.5-air

1. Project Overview

Problem Statement: Financial scams in Malaysia are escalating at an alarming rate. Victims lose an average of RM 10,000+ per incident, yet the reporting process is fragmented, slow, and emotionally traumatic. Victims must navigate multiple agencies (PDRM, BNM, NSRC, MCMC) with no single point of intake. Critical recovery windows, as short as 30 minutes for bank freeze requests are routinely missed due to process inefficiency.

Target Domain: Malaysian financial crime reporting, victim support, and law enforcement coordination.

Proposed Solution Summary: ScamBreaker is an AI-powered scam case management platform that enables victims to file reports conversationally in 4 languages, automatically classifies scam type, calculates fund recoverability score, generates formal documents (police report, bank dispute letter), and routes cases to the correct authorities are all within minutes of the incident. It bridges victims and enforcement agencies through a unified, intelligent intake pipeline.

2. Background & Business Objective

2.1 Background of the Problem

Malaysia recorded over 67,000 scam cases in 2023, with total losses exceeding RM 1.2 billion. The existing reporting pathway requires victims to physically visit police stations, call different hotlines per agency, and re-explain their situation multiple times in a state of distress. Evidence is scattered; screenshots, transaction histories, and chat logs are rarely compiled correctly. Authorities receive poorly structured reports that delay investigation and reduce recovery probability.

The problem is compounded by the multi-ethnic, multi-lingual nature of Malaysia. Many victims are not comfortable filing reports in Bahasa Melayu or English, particularly senior citizens and rural communities targeted by investment and parcel scams.

2.2 Importance of Solving This Issue

Bank freeze operations can only be initiated within 1–2 hours of a fraudulent transfer. After this window closes, fund recovery drops from ~40% to under 5%. A streamlined, AI-assisted intake tool directly impacts financial outcomes for victims. Reducing the reporting friction also improves case data quality, enabling authorities to detect scam networks and issue public warnings faster.

2.3 Strategic Fit / Impact

ScamBreaker aligns with Malaysia's National Scam Response Centre (NSRC) mandate and BNM's Financial Consumer Alert framework. By integrating GLM (Z AI Model) as the core reasoning engine, ScamBreaker demonstrates how AI can serve public safety infrastructure.

The platform is designed to be extensible to ASEAN-wide deployments where similar multi-language, multi-agency challenges exist.

3. Product Purpose

3.1 Main Goal of the System

To provide an AI-assisted, multi-lingual scam reporting and case management platform that minimises the time between incident and authority action, maximises fund recovery probability, and reduces emotional burden on victims.

3.2 Intended Users (Target Audience)

User Role	Description	Primary Need
Victim (Primary)	Malaysian resident who has been financially scammed	Fast, guided, empathetic intake in their preferred language
Authority Officer (Primary)	PDRM, NSRC, BNM, MCMC officer handling cases	Prioritised, structured case queue with recovery urgency
System Admin (Secondary)	Platform administrator managing agency routing	Agency configuration, monitoring, audit logs

4. System Functionalities

4.1 Description

ScamBreaker operates as a conversational AI intake engine. The system detects scam type from free-text input, extracts structured data (amount, bank, transfer method, timeline), calculates a real-time recoverability score, generates formal documents via GLM, and routes cases to the appropriate Malaysian authority. Authorities access a command centre dashboard with SLA countdown timers, duplicate detection, and workflow state management.

4.2 Key Functionalities

- Conversational Scam Intake: Victims describe their situation in natural language. The AI (GLM) asks targeted follow-up questions one at a time to complete the case profile.
- Multi-Language Support: Full support for English, Bahasa Melayu, Mandarin (Simplified), and Tamil; detected automatically from victim input.
- OCR Evidence Processing: Client-side OCR via Tesseract.js extracts text from uploaded screenshots and PDFs without uploading sensitive data to third-party servers.
- AI Case Analysis: GLM classifies scam type (e-commerce, investment, love, job, loan, impersonation, parcel, etc.), assigns urgency, priority, and summarises the case.
- Recoverability Engine: Rule-based scoring (0–100) evaluating 6 factors (transaction recency, transfer type, communication channel, financial severity, evidence quality, and mule account indicators). Score decays dynamically over time.

- Automated Document Generation: GLM generates a formal police report (Bahasa Melayu) and bank dispute letter (English) per case. Five document drafts are produced automatically.
- Authority Command Centre: Prioritised case dashboard with SLA countdown timers, workflow state machine, internal notes, duplicate detection, and audit logging.
- Agency Routing: Automatic routing to 30+ Malaysian agencies/banks based on scam type and keyword matching.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

ScamBreaker uses ZhipuAI GLM (Z AI Model) exclusively as mandated by UMHackathon 2026. Three model variants are deployed in a fallback chain: GLM-5.1 (primary, highest capability for classification and document generation), GLM-4.6v (secondary, vision-capable for evidence analysis), and GLM-4.5-air (tertiary, lightweight fallback). This ensures 99%+ availability and it's critical for a victim-facing emergency service.

4.3.2 Prompting Strategy

ScamBreaker employs a multi-step agentic prompting strategy across 4 distinct prompt types: (1) Conversational intake using a carefully engineered system prompt that instructs the model to detect language, classify scam type, extract structured data, and ask one follow-up question at a time. The model outputs a mandatory JSON block when trigger conditions A, B, or C are met (minimum data gathered). (2) Zero-shot classification for case analysis; structured JSON output at temperature 0.1 to ensure deterministic results. (3) Low-temperature (0.1) document generation prompts for police reports (BM) and bank dispute letters (EN) ensuring factual accuracy. (4) Statement restructuring prompt that normalises messy victim input into formal victim statements using the Malaysia bank reference table.

4.3.3 Context & Input Handling

The chat API receives the full conversation history on every turn to maintain context. Client-side OCR text is appended to the message payload before sending to the backend, keeping the context window populated without server-side file storage. Maximum input size is bounded by GLM's context window (128K tokens for GLM-5.1). For oversized inputs, the frontend truncates OCR text at 2,000 characters per file and concatenates up to 5 files, flagging truncation to the user.

4.3.4 Fallback & Failure Behaviour

fetchWithModelFallback() implements a 3-tier fallback: GLM-5.1 → GLM-4.6v → GLM-4.5-air. Each model receives 2 retries with exponential backoff (300ms × 2^n) and a 15-second AbortController timeout. If all 3 models fail, mockAnalyzeCase() returns a safe, pre-defined fallback result with a LOW urgency classification and a prompt for the victim to contact NSRC directly (1-800-886-688). The UI displays a graceful error state without exposing technical details.

5. User Stories & Use Cases

5.1 User Stories

- As a scam victim, I want to describe what happened in my own words (in BM/EN/ZH/TA) so that I do not need to navigate confusing forms while in distress.
- As a scam victim, I want to know my fund recovery chances immediately so that I can take urgent action before the recovery window closes.
- As a scam victim, I want a pre-filled police report and bank dispute letter generated for me so that I can file them without writing skills.
- As an authority officer, I want cases prioritised by recoverability urgency so that I can process CRITICAL cases first within SLA windows.
- As an authority officer, I want to see duplicate case alerts so that I can identify scam networks targeting multiple victims.
- As an authority officer, I want an audit trail of every action so that case handling is accountable and defensible.

5.2 Use Cases (Main Interactions)

UC-01: Victim Scam Report Submission

Victim opens ScamBreaker, selects language, describes the scam in the chat interface, uploads evidence (optional). AI asks clarifying questions. When sufficient data is collected, AI presents a summary card. Victim reviews and submits. System creates a structured case, calculates recoverability, generates documents, and routes to the appropriate agency.

UC-02: Authority Case Management

Authority officer logs in to the Command Centre. Cases are displayed sorted by recoverability (CRITICAL → HIGH → MEDIUM → LOW) with real-time SLA countdown. Officer opens a case, reviews AI-generated summary and documents, transitions workflow status, adds notes, or requests more information from the victim. All actions are audit-logged.

6. Features Included (Scope Definition)

#	Feature	Description	Priority
1	Conversational Intake (Chat UI)	AI-guided report filing in 4 languages with OCR evidence upload	P0 - Core
2	GLM Case Analysis & Classification	Automated scam type classification, urgency, priority, summary	P0 - Core
3	Recoverability Engine	Real-time 0–100 score with dynamic time decay and SLA countdown	P0 - Core
4	Automated Document Generation	Police report (BM) + bank dispute letter (EN) via GLM	P0 - Core
5	Authority Command Centre	Prioritised case queue, workflow state machine, duplicate detection	P0 - Core

#	Feature	Description	Priority
6	Agency Routing (30+ agencies)	Automatic routing to PDRM, BNM, NSRC, MCMC, banks	P1 - High
7	Victim Case Detail View	Victim can track case status and view generated documents	P1 - High
8	JWT Authentication (Victim & Authority)	Secure role-based access with HTTP-only cookie sessions	P0 - Core
9	Audit Logging (WorkflowLog)	Every system, victim, and authority action logged with timestamps	P1 - High
10	Multi-Model GLM Fallback	3-tier model fallback with exponential backoff and mock fallback	P1 - High

7. Features Not Included (Scope Control)

- Real-time notifications / SMS / email alerts to victims on case status updates
- Mobile native app (iOS/Android); web application only for this MVP
- Public-facing scam alert database or community reporting feed
- Integration with live bank APIs for automated freeze initiation
- Advanced analytics dashboard for national-level scam trend reporting
- Multi-user authority collaboration (shared editing); single-officer workflow only
- Victim identity verification (MyKad integration); manual IC number entry only
- Scheduled / asynchronous AI re-analysis of older cases

8. Assumptions & Constraints

8.1 LLM Cost Constraint

Estimated token cost per average user session (chat intake + case creation): ~12,000 tokens input + ~4,000 tokens output ≈ 16,000 tokens total. At GLM-5.1 pricing, this is approximately RM 0.08–0.12 per session. To manage inference costs: (1) the chat loop is capped at a maximum of 8 turns before forcing submission; (2) statement restructuring and document generation run only once per case (not re-generated unless explicitly requested); (3) temperature=0.1 for deterministic tasks reduces token waste from verbose outputs.

8.2 Technical Constraints

- The platform is a Next.js web application; no native mobile app is provided for this hackathon submission.
- OCR accuracy depends on image quality; blurry or handwritten documents may produce incomplete extractions.
- GLM API availability is subject to ZhipuAI infrastructure; the fallback chain mitigates but does not eliminate this dependency.
- PostgreSQL database is hosted locally/on a single server; no horizontal database scaling is implemented for this MVP.

8.3 Performance Constraints

- GLM API calls must complete within 15 seconds (AbortController timeout). Case creation pipeline (7 steps) may take 20–40 seconds for full completion.
- The authority dashboard polls every 30 seconds; real-time WebSocket push is not implemented in this version.

8.4 User Input

- The system requires at minimum: scam description, estimated amount lost, and transfer method to generate a valid case. Reports without these 3 elements will prompt the AI to request clarification before submission is enabled.

9. Risks & Questions Throughout Development

Risk	Question	Mitigation
GLM Hallucination in Documents	What if the generated police report contains fabricated details?	All GLM document prompts include explicit instructions to use only facts from the case record. Documents are clearly labelled DRAFT for human review before official use.
Recoverability Score Gaming	Can a victim manipulate the score by entering false timeline information?	Scores influence internal prioritisation only; victims do not see their own score. Authorities retain override capability.
OCR Privacy	Is client-side OCR sufficient to protect sensitive document data?	Tesseract.js runs entirely in the browser; raw image data never leaves the user's device. Only extracted text is sent to the server.
Multi-Agency Routing Accuracy	What if a case is routed to the wrong agency?	Routing is based on 30+ keyword rules. Authorities can manually re-route. The routing algorithm is transparent and auditable.
Emotional Distress UX	How does the chat UI avoid re-traumatising victims?	The system prompt instructs GLM to maintain an empathetic, non-judgmental tone. No accusatory language is ever used. The UI includes a crisis helpline link (NSRC 1-800-886-688).